ER. VISHAL SINHA

AI OPERATIONS | PYTHON AUTOMATION | APPLICATION & INFRA MANAGEMENT | SDH/DWDM NETWORK OPS

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PROFESSIONAL SUMMARY

Engineering and operations professional with 26 years of experience spanning in enterprise application infrastructure operations, telecom transmission network, and optical networking. Currently managing CATS and EPOLL applications at Bharti Airtel, focusing on service availability, server modernization, vulnerability remediation, monitoring, and audit readiness. Building practical capability in developing Python automation and generative AI capabilities through hands-on projects involving LLM chatbots, API-driven data extraction, reporting automation, and e-commerce platform development.

CORE COMPETENCIES

AI-assisted Operations • Python Scripting & Automation • Generative AI & Prompt Engineering • Application & Server Operations • Linux / RHEL Administration • Incident Management & Root-Cause Analysis • Vulnerability & Patch Management • IT Audits (GITC / BCMS) • Monitoring (Grafana / Zabbix) • Oracle SQL • SDH / DWDM / Microwave Networks • Stakeholder & Vendor Coordination

AI, PYTHON & AUTOMATION PROJECTS

AI Operations & Audit Copilot | Python, Streamlit, SQLite, FAISS, Hugging Face, AI/RAG

Developed AI-powered enterprise operations and risk management platform integrating CMDB inventory, vulnerability management, audit tracking, SOP knowledge retrieval, risk scoring, executive dashboards, and automated reporting. Implemented semantic search using Hugging Face embeddings and FAISS, conversational AI capabilities, application risk scoring, risk correlation analytics, and executive PDF reporting. Designed to provide centralized visibility into IT assets, security posture, audit observations, and operational procedures.

GitHub: https://github.com/vsinha2683/AI-Operations-Audit-Copilot.git

Telegram LLM Chatbot .

- Developed a Telegram chatbot using a lightweight LLM to automate customer support, answer product-related questions, and support customer engagement for an apparel e-commerce business.

GitHub: https://github.com/vsinha2683/telegram_LLM_Chatbot

YouTube Data Extraction & Reporting Automation.

- Build a Python-based analytics workflow using YouTube Data API v3, Google Sheets API, and Google Cloud Platform to retrieve video metadata and store structured results for reporting.
- Configured cloud APIs and service-account authentication; used REST APIs and JSON processing to automate collection and reduce manual reporting effort.

GitHub: https://github.com/vsinha2683/YouTube-Data-Scraper-Reporting-Automation

Saree & Suit E-commerce Platform.

Developed an e-commerce platform with user registration, profile management, OTP-based authentication, Razorpay payment integration, and an admin portal for product and inventory management.

- Implemented automated banner sliders, product galleries supporting up to five images and one video per product, image zoom, and admin password-reset functionality.

GitHub: https://github.com/vsinha2683/Full-featured-e-commerce-platform-for-sarees-and-suits-.git

PROFESSIONAL EXPERIENCE

Application Manager — CATS & EPOLL Applications | Bharti Airtel Ltd

July 2013 – Present

- Ensured **99.95% application availability** and SLA compliance for CATS & EPOLL platforms.
- Managed **30+ Linux servers**, including OS/DB upgrades, security patching, alarm extensions, and server hygiene activities.
- Achieved **95% vulnerability remediation** through package updates and security compliance initiatives.
- Executed **Linux 7 to Linux 8 migration** for 24 servers and built 5 new RHEL 8 database servers with seamless application cutover.
- Implemented and renewed **SSL certificates** for business-critical applications.
- Managed production user administration, password policies, DB links, and application connectivity.
- Created and refreshed **UAT environments** by cloning production instances for business testing.
- Configured and deployed 1**85+ cloud label printers** across Airtel warehouses for WMS/LPN printing.
- Conducted root cause analysis (RCA), preventive actions, and resolution of server alarms and incidents.
- Provisioned new servers and coordinated infrastructure enhancements with UNIX, DBA, Network, Security, and Application teams.
- Managed Service Requests, Firewall Policy Changes (PCRs), User Access Requests, and SOD approvals for business and project requirements.
- Led compliance activities including GITC, BCMS, vulnerability management, audit support, DRL submissions, and closure of audit observations.
- Integrated servers with enterprise monitoring platforms such as **Grafana and Zabbix**.
- Managed **EPOLL program implementation**, stakeholder coordination, issue resolution, escalation management, and project governance.
- Prepared BRDs, coordinated requirement discussions, supported UAT, provided deployment approvals, and performed post-production validations.
- Collaborated with vendors and support teams for resolution of NMS and application data issues.
- Managed ERP-CATS integration activities including BOM uploads, Item Master updates, functional classification alignment, and configuration of item attributes.
- Proficient in ServiceNow, Jira, Confluence, ARCOS, Cobalt, Grafana, Zabbix, and other enterprise support tools.

Assistant Manager — O&M Transmission | Bharti Airtel Ltd — Bhagalpur, Bihar

September 2010 – June 2013

- Ensured **99.95%+ network availability** through proactive monitoring, preventive actions, and field team coordination.
- Managed service continuity and uptime for **Voice, DSL, and SDH network infrastructure**.
- Monitored network performance and resolved outages to maintain SLA commitments.
- Prepared and managed annual **OPEX budgets**, ensuring spending remained within approved targets.
- Conducted expense analysis and implemented cost-optimization initiatives to improve operational efficiency.
- Identified and replicated best practices and innovative solutions to reduce operating costs.
- Developed and enforced SOPs for network operations, infrastructure protection, and asset management.
- Led field teams through regular reviews, guidance, and site visits to ensure operational excellence and compliance.
- Safeguarded **OFC, network equipment, infrastructure, and telecom sites** through preventive maintenance and governance measures.
- Coordinated with cross-functional teams to drive network reliability, service quality, and cost control initiatives.

Area Manager — O&M Transmission Networks | Reliance Communications Ltd — Rourkela, Orissa

November 2006 – August 2010

- Managed planned maintenance, hardware/software upgrades, bandwidth augmentation, and preventive maintenance activities across transmission networks.
- Performed **O&M of Microwave and SDH networks**, including live traffic migration, fault resolution, and service restoration with minimal downtime.
- Commissioned, configured, and tested **Nortel SDH transmission equipment** including OPTera LH 1600, TN16XE (STM-16), TN4XE (STM-4), and TN1C (STM-1).
- Executed **network upgrades and capacity expansions**, including STM-1 to STM-4 migrations, card expansions, and aggregate card replacements in live networks.
- Added and integrated **new DWDM wavelengths (lambdas)** into live networks, including network equalization and performance validation.
- Conducted stability testing, EOW testing, and acceptance testing using Acterna SDH Analyzers.
- Performed backup, upgrade, and restoration of live network elements while ensuring uninterrupted service continuity.
- Configured and tested **external alarms**, managed network element integration, and restored traffic following upgrades and maintenance activities.
- Monitored network performance and faults using **EC-1 NMS and Priside**, ensuring optimal network health and availability.
- Coordinated with operations teams for efficient fault management, preventive maintenance, and network reliability improvements.

Senior Engineer — Testing (SDH / DWDM / DLC) | HFCL — Solan

November 2003 – November 2006

- Led BSNL validation, TEC approval, and quality testing of STM-1/4/16 and DWDM equipment.
- Conducted BER testing and investigated alarms including LOS, LOF, AIS, LOP, RDI, REI, and B1/B2/B3.
- Performed card-level testing and repair and led installation/commissioning of Alcatel DWDM equipment across VSNL rings.

Customer Support Engineer | Telerex Communications India Pvt. Ltd. — New Delhi

March 2000 – October 2003

- Supported service provisioning and fault resolution for POTS, leased lines, E1 PRI, and ISDN BRI; prepared connectivity plans and commissioned Alcatel 4400 switches.
- Managed planned maintenance, hardware/software upgrades, bandwidth upgrades, and corporate customer issues.

Commissioning Engineer | Maruti Transmission (Franchisee of Hughes-Escorts Communication Ltd.)

January 1999 – February 2000

- Tested personal earth station units, adjusted antenna polarization and gain, commissioned VSAT terminals, conducted site surveys, and resolved post-installation issues.

TECHNICAL SKILLS

- AI & Productivity: ChatGPT (GPT-4), Claude, Gemini, GitHub Copilot; prompt engineering for coding, log analysis, incident triage, documentation, and operational runbooks.
- Programming & Web: Python, Java (Servlets, JSP, JDBC, RMI, JavaBeans), C/C++, JavaScript, HTML, CGI/Perl, Fortran 77, Pascal.
- Data & Databases: Oracle Database, SQL, JSON, REST APIs, Google Sheets API, YouTube Data API v3.
- Systems & Tools: Linux administration, RHEL 7/8, Windows NT/95/98/MS-DOS, Grafana, Zabbix, ServiceNow, Jira, ARCOS, Cobalt, Confluence.
- Telecom & Optical: SDH (STM-1/4/16), DWDM, NEC PDH/SDH microwave, DLC, OTDR, OSA, DTA, CDMA testers, E1/BER testers.
- Governance & Security: Vulnerability closure, patch management, SSL certificates, SoD approvals, BISP policy overrides, GITC and BCMS audits.

EDUCATION & TRAINING

Bachelor of Engineering (Electronics & Communication) — Karnataka University, Dharwad | 1998 | First Class

- CDMA-WLL BSC & BTS — Huawei professional training.
- GSM — Nokia professional training.

- DWDM — Alcatel professional training.
- PDH/SDH NEC Microwave — Reliance professional training.
- Generative AI & Prompt Engineering — practical proficiency with ChatGPT, Claude, Gemini, GitHub Copilot, VS Code, Jupyter, and Google Colab.

SELECTED PROGRAMS & PROJECTS

- EPOLL Modernization: Coordinated BRD, UAT, production deployment, post-deployment validation, and stakeholder communication.
- Server Modernization: Led 24 RHEL 7-to-RHEL 8 in-place upgrades and five new RHEL 8 database server builds.
- Pan-India WMS Printer Rollout: Configured and tested 200+ cloud label printers for warehouse LPN printing.
- Alcatel DWDM Deployment: Led installation and commissioning across multiple rings through VSNL acceptance testing.

ACHIEVEMENTS

- Awarded Performance Linked Incentive (PLI) during the Mid-Term KRA Assessment for successfully leading critical audit closures, ensuring compliance adherence, and driving operational excellence.
- Received Employee of month Award for outstanding performance and contribution in timely RHEL upgrade &Vulnerability closure.

ADDITIONAL INFORMATION

Current location: Delhi NCR | Open to relocation

INTERESTS

Continuous learning and professional development in modern technologies and digital transformation, Ghazal, Current Affairs

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